Lesson notes · for you

Your data, your bundle

AI agents · Lower secondary · Lesson 4

Time
20 min

Materials
Your phone's data usage screen, or a family phone's with the owner's permission; A notebook for a week's data diary; A pen

Learning goals

- Compare a model on your phone with one in the cloud.
- Work out a month's data with a short program.
- Decide what you will never type into a cloud agent.

Activity: Phone or cloud?

1. Warm-up (3 min): Guess how much data your phone (or your family's) uses in a month, then check the data usage screen if you can.
2. Work it out (6 min): One cloud request is about 18 KB. Work out a month at 5, 20 and 50 requests a day, and compare each with Darsi Tiny's 2.8 MB, downloaded once.
3. Sort (8 min): Write ten questions you might ask an AI (homework, a diary entry, a friend's photo, the weather). Sort each into: fine on the phone, fine in the cloud, or never type it.
4. Rules (3 min): Write your own three rules, and share one with a partner.

A phone helps but isn't needed: the sums and the sorting work on paper. Never use anyone's real private details as examples.

Discussion

- When is a big cloud model worth its cost?
- Why might a free AI app still cost you something?
- What would you tell a younger student about typing into chatbots?

Check yourself

- Write one question you'd ask on the phone, one in the cloud, and one thing you'd never type.

Try it this week
Check how much data your phone used last month and on which apps. Then write your own three rules: what you'll ask on the phone, what in the cloud, and what you'll never type into any agent.

Worksheet

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Name _____ Class _____ Date _____

1 **You want help with your diary, which is full of private thoughts. Where is it safest to ask?**

◯ A model on your phone, or nobody ◯ A cloud chatbot: it knows more

◯ A cloud chatbot, in a new chat

2 **Which of these should you never type into a cloud agent? Choose all that apply.**

☐ A question about how volcanoes form ☐ Your home address ☐ Your password

☐ A fractions problem from your textbook ☐ A friend's exam marks

3 **Complete the program so it works out a month of cloud requests in megabytes.**

```
kb_each = 18
per_day = 20
days = 30
kb = kb_each _____ per_day * days
print(kb _____ 1000, "MB")
```

Word bank: `+` `*` `-` `/`

4 **What does this program print?**

```
tiny = 2.8          # MB, once
cloud = 18 / 1000   # MB a request
print(round(tiny / cloud))
```

◯ 28 ◯ 156 ◯ 0.018

5 **A small model on your phone knows as much as a big cloud model.**

◯ True ◯ False

✎ **Which AI apps do you or your family use? For each, would you rather it ran on the phone or in the cloud, and why?**

Check your answers

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1 **A model on your phone, or nobody**

Whatever you type into a cloud chatbot leaves your phone, even in a new chat.

2 **Your password; A friend's exam marks; Your home address**

Passwords, your address and other people's private things should never leave your phone.

3

```
kb_each = 18
per_day = 20
days = 30
kb = kb_each * per_day * days
print(kb / 1000, "MB")
```

Multiply 18 KB by 20 requests a day and by 30 days: 10,800 KB.

4 **156**

One cloud request is 18 / 1000 = 0.018 MB, and 2.8 / 0.018 is about 156.

5 **False**

Darsi Tiny knows ten school tools and almost nothing else; a big cloud model knows far more.